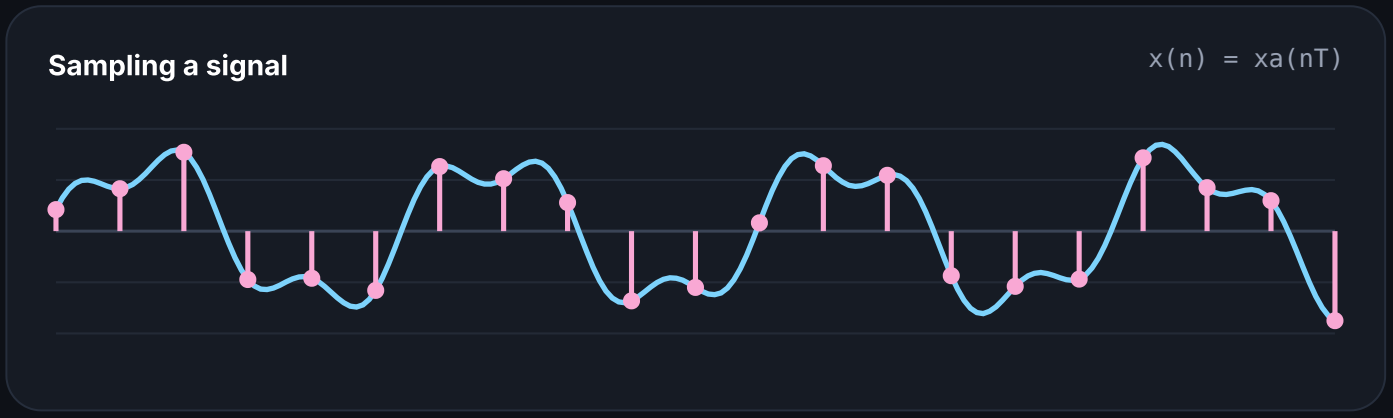
النظري
Chapter 2

Digital Signal Processing · Theory · 13 questions



DFT IDFT Parseval FFT Butterfly IFFT

01 محاضرة 5 — 8

02 محاضرة 6 — 5

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Modern Academy for Engineering & Technology

DSP / ch02 / contents

المحتوى

Theory · 13 questions

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DFT, IDFT & Properties
8 سؤال p. 4

02 **محاضرة 6**
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إزاي تذاكر من الملف

الإجابات بجُمل الدكتور. جوه كل محاضرة الأسئلة مترتبة من الأكتر تكرارًا في الامتحانات. المربع الأخضر (الفكرة) هو الإجابة في سطر، و Key words تسمّع منها وانت مغطي الإجابة. في آخر كل محاضرة Cheat sheet.

DSP / ch02 / repeated

الأسئلة اللي بتتكرر أكتر (ابدأ بيها)

Most repeated in previous exams — start with these

السؤال	عدد مرات وروده
Q3. Prove that the DFT, X(k), of a sequence x(n) is periodic.	13
P9. x(n) = {1, 0, 1, 1} — DFT / FFT / IFFT / Parseval / spectra / saving	10
P10. x(n) = {1, 0, 1, 0} — DFT / FFT / IFFT / Parseval / spectra / saving	4
P12. x(n) = {1, 0, 0, 1} — DFT / FFT / IFFT / Parseval / spectra / saving	3
Q8. Check (✓) or (×) and correct the mistakes (DFT / FT statements).	3

SECTION 01

محاضرة 5

DFT, IDFT & Properties

الأسئلة — الأكثر تكرارًا الأول

p. 5	Prove that the DFT, X(k), of a sequence x(n) is periodic.	Q3 ☐
p. 6	Check (✓) or (×) and correct the mistakes (DFT / FT statements).	Q8 ☐
p. 7	Define the Discrete Fourier Transform (DFT).	Q1 ☐
p. 8	State the mathematical properties of the DFT.	Q6 ☐
p. 9	Find the relation between the DFT and the Fourier transform.	Q2 ☐
p. 10	Show that the amplitude spectrum of an N-point DFT is symmetrical about harmonic N/2...	Q4 ☐
p. 11	Define the Inverse Discrete Fourier Transform (IDFT).	Q5 ☐
p. 11	State Parseval's theorem.	Q7 ☐

محاضرة 5

×13 متكرر Q3

Prove that the DFT, X(k), of a sequence x(n) is periodic.

$$X(k) = \Sigma_{n=0}^{N-1} x(nT)\ e^{-j2\pi kn/N}$$

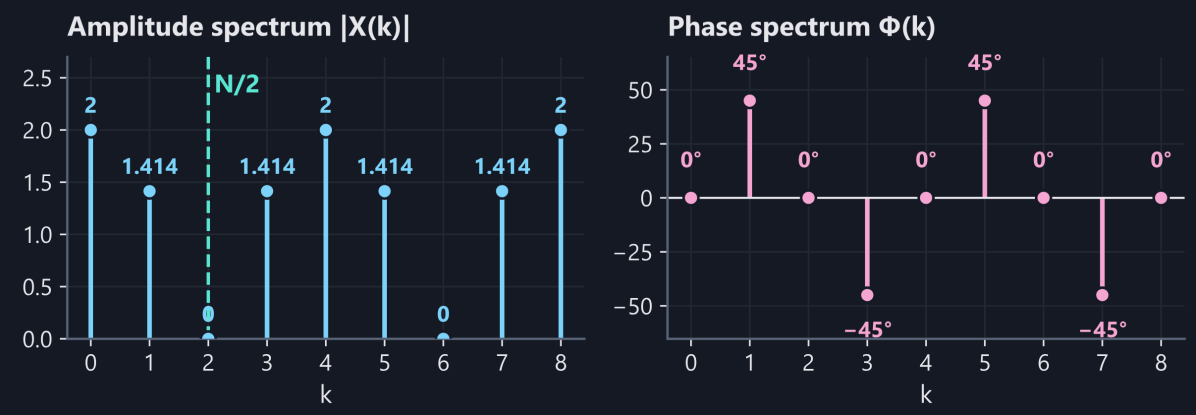
$$X(k+N) = \Sigma_{n=0}^{N-1} x(nT)\ e^{-j2\pi kn/N} \cdot e^{-j2\pi Nn/N} = \Sigma_{n=0}^{N-1} x(nT)\ e^{-j2\pi kn/N} \cdot e^{-j2\pi n}$$

Since n is integral, $e^{-j2\pi n} = 1$, so:

$$X(k+N) = X(k)$$

The DFT is periodic with period N (the cyclical property of the DFT). E.g. for {1, 0, 0, 1}: X(0) = 2, therefore X(4) = 2 also.

|X(k)| for {1, 0, 0, 1}: X(k + 4) = X(k), symmetric about N/2



الفكرة e^{−j2πn} = 1 for integer n, so X(k + N) = X(k): the DFT repeats every N.

المصدر — من الـ PDFs بتاعتك

المحاضرة **LEC 5** · Sec. 2.1 · p. 4
1-Lectures/Lectures Fall 2026/Chapter 2/DSP-LEC-5-Ch-2.pdf

جه في
Final 7-6-2023 · Q1-b | Final 9-6-2026 · Q2-a | Final 17-8-2026 · Q2-a
Final 23-1-2023 · Q3-a | Final 12-1-2019 · Q2-b | Final 2-1-2017 · Q2-b
Final 23-12-2017 · Q2-b | Midterm 12-11-2025 · Q2-a | Midterm 28-11-2021 · Q2-a
Midterm 8-12-2020 · Q2-b | Midterm 1 Model-2 · Q2-a | Midterm 2 · Q2-b

محاضرة 5

Sheet 2 · 2.1

2-Section/Sheets/ · 5-Exams/Final/ · 5-Exams/Midterm/

×3 **Q8**

Check (✓) or (×) and correct the mistakes (DFT / FT statements).

1. Fourier transform (FT) is calculated for non-periodic continuous-time signals. **(✓)**
2. Discrete Fourier transform (DFT) is obtained for the analog signals. **(×)**
   Correct: the DFT is obtained for the sampled (discrete-time) sequence x(nT).

الفكرة FT → non-periodic continuous-time signals (✓); DFT → sampled sequence x(nT), not the analog signal (× as written).

المصدر — من الـ PDFs بتاعتك

المحاضرة **LEC 5** · Sec. 2.1 · p. 4

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جه في Final 12-1-2019 · Q5 | Final 23-12-2017 · Q5-b | Quiz 1 Summer 26 · Q3

5-Exams/Final/ · 5-Exams/OLD Quiz/

محاضرة 5

×1 **Q1**

Define the Discrete Fourier Transform (DFT).

Assume a waveform has been sampled at regular time intervals T to produce the sample sequence {x(nT)} = x(0), x(T), ..., x[(N − 1)T] of N sample values. The DFT of x(nT) is defined as the sequence of complex values {X(kΩ)} = X(0), X(Ω), ..., X[(N − 1)Ω] in the frequency domain, where Ω = 2π/NT is the first harmonic frequency:

$$X(k) = \Sigma_{n=0}^{N-1} x(nT)\, e^{-jk\Omega nT}, \quad k = 0, 1, \ldots, N-1 \quad (2.4)$$

$$X(k) = R(k) + jI(k) \quad (2.1) \qquad |X(k)| = [R^2(k) + I^2(k)]^{\frac{1}{2}} \quad (2.2) \qquad \Phi(k) = \tan^{-1}[I(k)/R(k)] \quad (2.3)$$

الفكرة X(k) = Σ x(nT) e^{−jkΩnT}, k = 0 ... N − 1: N samples in time give N complex frequency components spaced Ω = 2π/NT.

المصدر — من الـ PDFs بتاعتك

المحاضرة

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جه في

Book exam Midterm 2015

7-Books (References)/Extract Question From Book/

محاضرة 5

×1 Q6

State the mathematical properties of the DFT.

1. **Symmetry:** Re[X(N − k)] = Re X(k) (2.8) and Im[X(N − k)] = −Im[X(k)] (2.9).
2. **Even functions:** if $x_e(n) = x_e(-n)$: $X_e(k) = \Sigma\, x_e(n) \cos(k\Omega nT)$ (2.10).
3. **Odd functions:** if $x_o(n) = -x_o(-n)$: $X_o(k) = -j\, \Sigma\, x_o(n) \sin(k\Omega nT)$ (2.11).
4. **Parseval's theorem:** $\Sigma\, x^2(n) = (1/N)\, \Sigma\, |X(k)|^2$ (2.12).
5. **Delta function:** $F_D[\delta(nT)] = 1$ (2.13).
6. Linear cross-correlation of two sequences may be computed using DFTs.
7. DFTs may be used to compute circular convolutions and, using augmenting zeros, linear convolutions.

الفكرة

Symmetry, even and odd functions, Parseval's theorem, delta function, linear cross-correlation and circular convolution.

Key words: Symmetry | Even functions | Odd functions | Parseval's theorem

المصدر — من الـ PDFs بتاعتك

المحاضرة

LEC 5 · Sec. 2.3 · p. 27
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جه في

Final 18-1-2024 · Q2-a
`5-Exams/Final/`

محاضرة 5

Q2 أقل تكرارًا

Find the relation between the DFT and the Fourier transform.

Equation (2.4) is analogous to the Fourier transform $F(j\omega) = \int f(t)\, e^{-j\omega t}\, dt$ (F-1). Putting $x(nT) = f(t)$, $k\Omega = \omega$, $nT = t$, $dt = T$, and replacing the integral by a summation (for $f(t) = 0$ outside $0 \le t \le (N-1)T$):

$$\Sigma_{n=0}^{N-1}\, x(nT)\, e^{-jk\Omega nT}\, T = F(j\omega) \quad (2.5)$$

$$\mathbf{F(j\omega) = X(k)\, T} \quad \text{i.e.} \quad \mathbf{FT = DFT \times T} \quad (2.6)$$

The FT components are obtained by multiplying the DFT components by the sampling interval T. Similarly, **IFT = IDFT / T**.

الفكرة Sampling the Fourier-transform integral gives $F(j\omega) = X(k)\cdot T$, i.e. FT = DFT × T (and IFT = IDFT / T).

المصدر — من الـ PDFs بتاعتك

المحاضرة

LEC 5 · Sec. 2.1 · p. 4

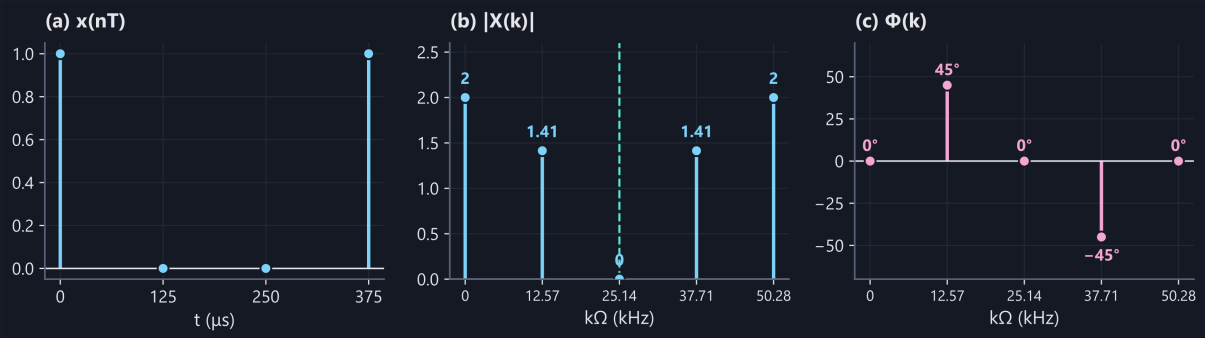
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محاضرة 5

أقل تكرارًا **Q4**

Show that the amplitude spectrum of an N-point DFT is symmetrical about harmonic N/2 (folding frequency).

- The amplitude spectrum of an N-point DFT is symmetrical about harmonic N/2, and the phase angles are an odd function centred round this component.
- If $2f_{max}$ samples per second are taken for t seconds, then $2f_{max}t = N$, and the first harmonic frequency is $1/t = 2f_{max}/N$.
- The symmetry at harmonic N/2 occurs at the frequency $(N/2)(2f_{max}/N) = \mathbf{f_{max}}$, the maximum frequency present in the signal (the **folding frequency**).
- So it is unnecessary to plot beyond N/2: N real data values transform to N/2 complex DFT values of practical significance.



الفكرة

For real x(n), X(N − k) = X*(k), so |X(k)| is symmetric about harmonic N/2 (the folding frequency Fs/2).

المصدر — من الـ PDFs بتاعتك

المحاضرة

LEC 5 · Sec. 2.1 (folding at N/2) · p. 19

`1-Lectures/Lectures Fall 2026/Chapter 2/DSP-LEC-5-Ch-2.pdf`

محاضرة 5

Q5 أقل تكرارًا

Define the Inverse Discrete Fourier Transform (IDFT).

To obtain the discrete transformation from the frequency to the time domain, the IDFT is used:

$$x(nT) = F_D^{-1}[X(k)] = (1/N) \sum_{k=0}^{N-1} X(k)\, e^{jk\Omega nT}, \quad n = 0, 1, \ldots, N-1 \quad (2.7)$$

The inverse Fourier transform is obtained from the IDFT by dividing by T: **IFT = IDFT / T**.

الفكرة x(nT) = (1/N) Σ X(k) e^{jkΩnT}: same form as the DFT with +j and a factor 1/N.

المصدر — من الـ PDFs بتاعتك

المحاضرة **LEC 5** · Sec. 2.2 · p. 21
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Q7 أقل تكرارًا

State Parseval's theorem.

The normalized energy in the signal is given by either of the expressions:

$$\sum_{n=0}^{N-1} x^2(n) = (1/N) \sum_{k=0}^{N-1} |X(k)|^2 \quad (2.12)$$

The left-hand side is the sum of the squared magnitudes of the time series; the right-hand side is the mean square spectral amplitude.

الفكرة Σ x²(n) = (1/N) Σ |X(k)|²: the energy is the same in time and in frequency.

المصدر — من الـ PDFs بتاعتك

المحاضرة **LEC 5** · Sec. 2.3 · p. 27
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محاضرة 5

★ Cheat sheet · محاضرة 5

افتكر

X(k) = Σ x(nT) e^{−jkΩnT}, k = 0 … N − 1: N samples in time give N complex frequency components spaced Ω = 2π/NT. **Q1**

Sampling the Fourier-transform integral gives F(jω) = X(k)·T, i.e. FT = DFT × T (and IFT = IDFT / T). **Q2**

e^{−j2πn} = 1 for integer n, so X(k + N) = X(k): the DFT repeats every N. **Q3**

For real x(n), X(N − k) = X*(k), so |X(k)| is symmetric about harmonic N/2 (the folding frequency Fs/2). **Q4**

x(nT) = (1/N) Σ X(k) e^{jkΩnT}: same form as the DFT with +j and a factor 1/N. **Q5**

Symmetry, even and odd functions, Parseval's theorem, delta function, linear cross-correlation and circular convolution. **Q6**

Σ x²(n) = (1/N) Σ |X(k)|²: the energy is the same in time and in frequency. **Q7**

FT → non-periodic continuous-time signals (✓); DFT → sampled sequence x(nT), not the analog signal (× as written). **Q8**

المعادلات

$$X(k) = R(k) + jI(k) \quad (2.1) \qquad |X(k)| = [R^2(k) + I^2(k)]^{\frac{1}{2}} \quad (2.2) \qquad \Phi(k) = \tan^{-1}[I(k)/R(k)] \quad (2.3)$$

$$X(k) = \sum_{n=0}^{N-1} x(nT)\, e^{-jk\Omega nT}, \quad k = 0, 1, \ldots, N-1 \quad (2.4)$$

$$\mathbf{F(j\omega) = X(k)\ T} \quad \text{i.e.} \quad \mathbf{FT = DFT \times T} \quad (2.6)$$

$$\sum_{n=0}^{N-1} x(nT)\, e^{-jk\Omega nT}\, T = F(j\omega) \quad (2.5)$$

$$\sum_{n=0}^{N-1} x^2(n) = (1/N) \sum_{k=0}^{N-1} |X(k)|^2 \quad (2.12)$$

$$x(nT) = F_D^{-1}[X(k)] = (1/N) \sum_{k=0}^{N-1} X(k)\, e^{jk\Omega nT}, \quad n = 0, 1, \ldots, N-1 \quad (2.7)$$

SECTION 02

محاضرة 6

FFT, Butterfly & IFFT

الأسئلة

p. 13 Discuss the computational complexity of the DFT. Q9 ☐

p. 15 Explain the decimation-in-time (DIT) FFT algorithm. Q10 ☐

p. 16 What is the butterfly? Draw it. Q11 ☐

p. 18 What are the computational advantages (saving) of the FFT? Q12 ☐

p. 19 Show how the IFFT is obtained from the FFT. Q13 ☐

Q9

Discuss the computational complexity of the DFT.

For an 8-point DFT (Eq. 2.15): $X(k) = x(0)e^{-jK0} + x(1)e^{-jK1} + \ldots + x(7)e^{-jK7}$, $k = 0, \ldots, 7$.

- Each X(k) needs 8 complex multiplications and 7 complex additions; for N points: N and N − 1.
- There are N harmonic components to evaluate (k = 0 ... N − 1).

Operation	DFT
No. of complex multiplications	N^2
No. of complex additions	N(N − 1)

If N = 1024, approximately one million complex multiplications and one million complex additions are required, so some means of reducing these numbers is desirable.

الفكرة

Direct DFT needs N^2 complex multiplications (N(N − 1) additions), which grows very fast with N.

محاضرة 6

المصدر — من الـ PDFs بتاعتك

المحاضرة

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1-Lectures/Lectures Fall 2026/Chapter 2/DSP-LEC-5-Ch-2.pdf

Q10

Explain the decimation-in-time (DIT) FFT algorithm.

- The FFT uses the computational redundancy in the DFT to reduce the number of calculations (Cooley and Tukey, 1965). Decimation refers to the reduction in calculations performed on time-domain data.
- Writing $W_N = e^{-j2\pi/N}$ (2.17), the DFT becomes $X(k) = \Sigma\, x_n\, W_N^{kn}$ (2.18), with:

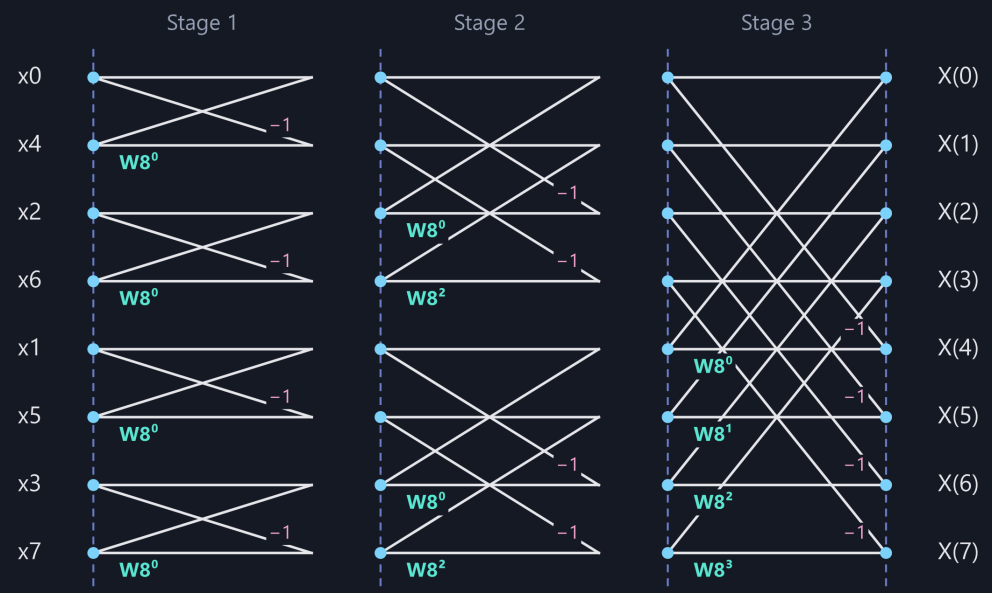
$$W_N^{2} = W_{N/2} \quad (2.21b) \qquad W_N^{k+N/2} = -W_N^{k} \quad (2.21c)$$

- The data are split into the even sequence x_{2n} and the odd sequence x_{2n+1} (N/2 points each):

$$X1(k) = X11(k) + W_N^{k}\, X12(k) \quad (2.24)$$

- X11(k) is the DFT of the even sequence and X12(k) of the odd sequence. Repeating the decomposition, an 8-point DFT becomes four 2-point DFTs (Table 2.1).

8-point DIT FFT: N/2 butterflies × $\log_2 N$ stages (Fig. 2.3)



محاضرة 6

الفكرة Split x(n) into even and odd samples repeatedly ($\log_2 N$ stages) until only 2-point DFTs remain, then combine with W_N^k.

المصدر — من الـ PDFs بتاعتك

المحاضرة

LEC 6 · Sec. 2.5 · p. 6

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Q11

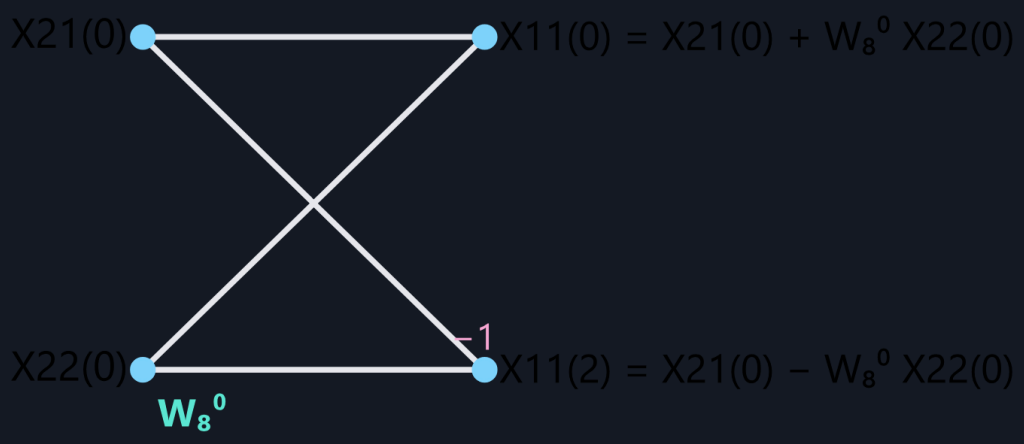
What is the butterfly? Draw it.

Equations 2.25 are represented diagrammatically by taking them in pairs, exploiting the symmetry centred on the sign differences:

$$X11(0) = X21(0) + W_8^0\ X22(0) \qquad X11(2) = X21(0) - W_8^0\ X22(0)$$

The inputs are at the left of the cross and the outputs at the right. The structure is called a **butterfly**, being reminiscent of the symbolic representation of the insect (Fig. 2.2).

The butterfly (Fig. 2.2 a)



محاضرة 6

الفكرة

Butterfly: X11(0) = X21(0) + W X22(0) and X11(2) = X21(0) − W X22(0) — one multiplication, two additions.

المصدر — من الـ PDFs بتاعتك

المحاضرة

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Q12

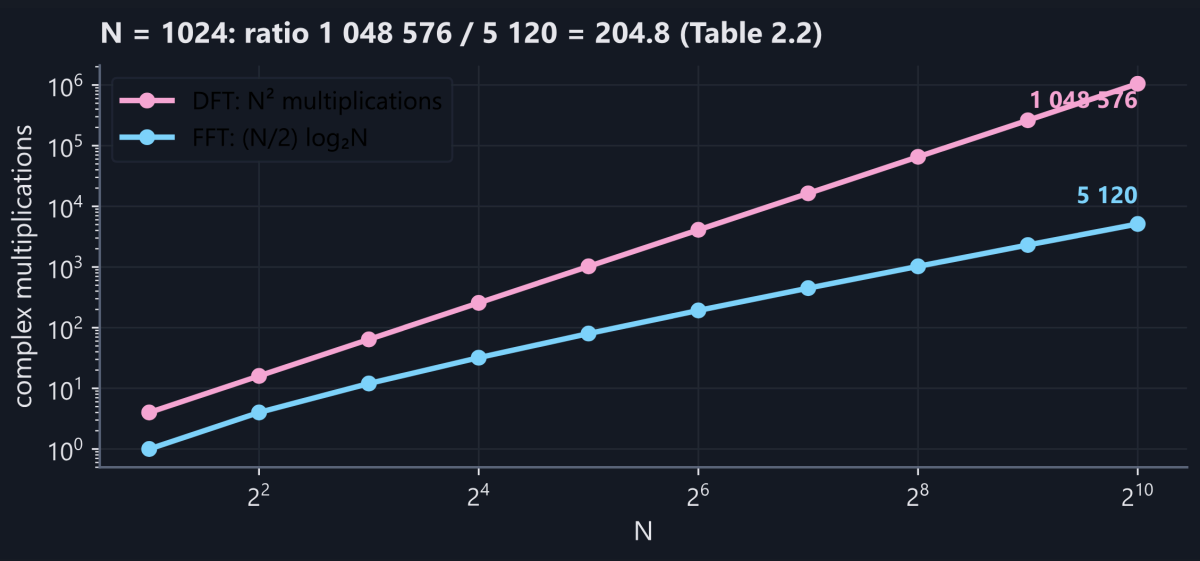
What are the computational advantages (saving) of the FFT?

- An N-point FFT contains N/2 butterflies per stage and $\log_2 N$ stages: $(N/2)\log_2 N$ butterflies.
- Each butterfly involves one complex multiplication and two complex additions.

	DFT	FFT	Saving
Complex multiplications	N^2	$(N/2)\log_2 N$	$N^2 - (N/2)\log_2 N$
Complex additions	$N(N-1)$	$N \log_2 N$	$N(N-1) - N\log_2 N$

N	DFT mult.	DFT add.	FFT mult.	FFT add.
4	16	12	4	8
8	64	56	12	24
1024	1 048 576	1 047 552	5 120	10 240

For N = 1024 the number of multiplications is reduced by a factor of 204.8.



الفكرة

FFT needs (N/2) $\log_2 N$ multiplications and N $\log_2 N$ additions instead of N^2 — e.g. 12 vs 64 for N = 8.

المصدر — من الـ PDFs بتاعتك

المحاضرة

LEC 6 · Sec. 2.5.2 · p. 18

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Q13

Show how the IFFT is obtained from the FFT.

Substituting in Eq. 2.4 (sum over λ, k → μ, Ω = 2π/NT):

$$\text{FFT:} \quad X(\mu) = \sum_{\lambda=0}^{N-1} x(\lambda)\, e^{-j\mu(2\pi/N)\lambda} \quad (2.26)$$

and similarly in Eq. 2.7 (k = λ, n = μ):

$$\text{IFFT:} \quad x(\mu) = (1/N) \sum_{\lambda=0}^{N-1} X(\lambda)\, e^{j\mu(2\pi/N)\lambda} \quad (2.27)$$

The IFFT differs from the FFT only in the factor **1/N** and the **sign of the exponent**, so with small modifications the FFT may be used to calculate the IFFT.

Procedure: re-order X(k) as in the FFT, use the same butterflies with W_N^{-k} instead of W_N^{k}, then divide the outputs by N.

الفكرة

Use the same flowgraph with W^{−k} instead of W^{k} and divide the result by N.

Key words Procedure

المصدر — من الـ PDFs بتاعتك

المحاضرة

LEC 6 · Sec. 2.6 · p. 20

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محاضرة 6

★ Cheat sheet · محاضرة 6

افتكر

Direct DFT needs N^2 complex multiplications ($N(N-1)$ additions), which grows very fast with N. **Q9**

Split x(n) into even and odd samples repeatedly ($\log_2 N$ stages) until only 2-point DFTs remain, then combine with W_N^k. **Q10**

Butterfly: X11(0) = X21(0) + W X22(0) and X11(2) = X21(0) − W X22(0) — one multiplication, two additions. **Q11**

FFT needs $(N/2)\log_2 N$ multiplications and $N\log_2 N$ additions instead of N^2 — e.g. 12 vs 64 for N = 8. **Q12**

Use the same flowgraph with W^{−k} instead of W^{k} and divide the result by N. **Q13**

المعادلات

$$X1(k) = X11(k) + W_N^k \, X12(k) \quad (2.24)$$

$$W_N^2 = W_{N/2} \quad (2.21b) \qquad W_N^{k+N/2} = -W_N^k \quad (2.21c)$$

$$\text{IFFT:} \quad x(\mu) = (1/N) \sum_{\lambda=0}^{N-1} X(\lambda) \, e^{j\mu(2\pi/N)\lambda} \quad (2.27)$$

$$\text{FFT:} \quad X(\mu) = \sum_{\lambda=0}^{N-1} x(\lambda) \, e^{-j\mu(2\pi/N)\lambda} \quad (2.26)$$